Personal Statement — MSc in Robotics and Autonomous Systems, University of Macau

Over the past two years, I have worked on three robots whose control had to be validated on real hardware: a RoboMaster hero robot for which I implemented the gimbal, launcher and chassis control loops; a wheel-legged robot that stays upright while its legs change length; and a three-omni-wheel ballbot that keeps its balance while two arms move on top of it. These projects led me to a clear conclusion: I can implement a controller and make it run, but I now need a more rigorous understanding of why it works and how to design it systematically. That is why I am applying to the MSc in Robotics and Autonomous Systems at the University of Macau.

My undergraduate study at Shenzhen Technology University (Computer Science, ranked 6 of 156) gave me a solid base in programming and algorithms. As my robotics projects increasingly required a stronger understanding of control, I also took Principles of Automatic Control and Modern Control Theory. During my internship at the Shenzhen Institute of Artificial Intelligence and Robotics for Society (AIRS), I applied this preparation to a real balance-control problem. I was responsible for the balance control of a dual-arm ballbot. I built and deployed a cascaded PID controller, then modelled the system as a four-state inverted pendulum and used LQR to analyse how state weights trade stability against control effort. The model explained tuning behaviour and actuator saturation that hand tuning alone could not. Compensating arm-induced centre-of-mass shifts with URDF/Pinocchio gravity feedforward showed me how tightly upper-body motion and base balance are coupled. I also integrated the ROS state processing with CAN motor execution and added state initialisation, output limiting and fall-recovery safeguards, so that the loop could be tested repeatedly without damaging the robot. The chassis ultimately maintained dynamic balance while both arms moved along fixed trajectories.

I then extended this control-oriented approach through two laboratory projects, moving from balance control to locomotion and navigation. On a variable-leg-length wheel-legged robot I implemented gain-scheduled LQR and virtual model control with closed-chain leg kinematics, and tuned it on an STM32H7/FDCAN platform. On a car-like mobile robot I built the full chain from EKF/AMCL localisation through Hybrid-A* and TEB planning to MPC tracking, and ran it on the real vehicle. Internships at Future Power and Kaihong then extended my experience in two directions: VR teleoperation of a dual-arm robot, where constrained inverse kinematics keeps the arms tracking the operator without velocity spikes near singularities, and multi-robot platform software on OpenHarmony.

The MSc in Robotics and Autonomous Systems at the University of Macau fits the breadth of robotics I now want to consolidate. My projects have covered balance control, legged locomotion, localisation, planning, trajectory tracking, manipulation and edge AI, but these experiences were developed on different platforms. A programme that brings together machine vision, sensor fusion, robot pose estimation, SLAM, nonlinear motion estimation and control, and trajectory planning would help me understand the common principles behind these systems. I am particularly interested in the connection between autonomous perception and control: a planner is only useful when the state estimate is trustworthy and the controller can execute the resulting trajectory under physical constraints.

After graduation I intend to work as a robotics algorithm engineer in the Greater Bay Area, focusing on autonomous navigation and control for mobile, legged and service robots. I would bring practical hardware experience and a habit of validating complete loops, while using the programme to strengthen my mathematical modelling, sensor-fusion and autonomous-system design skills.